

Week 3

Drag, lift, and force coefficients

Automotive Aerodynamics Independent Study

Industrial Design × Physics (DePaul)

Physics mentor · 10-week IS

Today's agenda

1. Recap: $\rho, q = KE/V$, and $F \sim \Delta P A$
2. Split aero force into drag and lift
3. Sign convention: downforce is negative lift
4. Derive $F = qCA$ from $\Delta P A$
5. Anatomy of each symbol in $F_D = \frac{1}{2} \rho v^2 C_D A$
6. Reference area A — pick one and freeze it
7. Worked checkpoint (including efficiency)
8. Order-of-magnitude C_D
9. CAD v1: body + wing, record AoA
10. Deliverables

Learning targets (by end of Week 3)

You will be able to:

- Derive $F = qCA$ from $F \sim \Delta PA$ with $\Delta P = Cq$
- Write and use $F_D = \frac{1}{2}\rho v^2 C_D A$ and $F_L = \frac{1}{2}\rho v^2 C_L A$, naming each symbol
- Treat **downforce as negative lift** (one convention for the term)
- Estimate order-of-magnitude forces for the scale model
- Define a **single reference area** A for comparing configs

Recap — Weeks 1 and 2

Week 1: $N = mg + D_w$, $F_{\text{drive}} = F_D$, forces from the CM.

Week 2: density $\rho = m/V$ (unit kg/m^3) turns a blob's kinetic energy into a pressure scale

$$q \equiv \frac{KE}{V} = \frac{1}{2} \rho v^2$$

Net aero force is about ΔP , and a first estimate was $F \sim qA$.

Today we insert the missing numbers C_D and C_L so different *shapes* can be compared at the same q and A .

Learning outcome LO3 (syllabus)

Predict and measure drag and lift/downforce coefficients for modular parts.

You cannot measure C_D until Week 6–7. You **can** already:

- Predict a plausible range
- Compute F_D, F_L from assumed C_D, C_L
- Freeze A so later measurements are comparable

Week 5 will warn that C_D at model Re is not automatically the full-car C_D .

Drag and lift are components

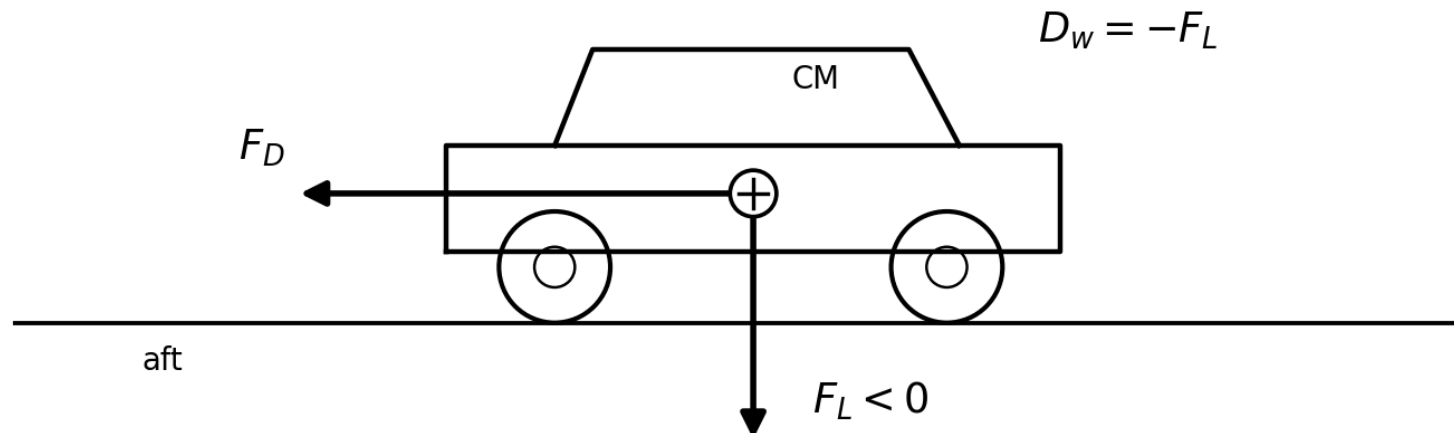
Relative to the free-stream direction:

- **Drag** F_D : parallel to the oncoming flow (resists motion; aft on the car)
- **Lift** F_L : perpendicular to the oncoming flow (aircraft sense: **up** positive)

Tails at the CM (Week 1 particle FBD). F_L up-positive, so downforce is $D_w = -F_L$.

Particle FBD: two aero components

Aero force has two components: drag and lift



Particle FBD (Week 1): tails at CM. F_D parallel to motion (aft). F_L up-positive.

Course sign convention (freeze this)

$$D_w = -F_L \quad \text{when } F_L \text{ is up-positive}$$

If $C_L = -0.30$, the vertical aero force is **down** (downforce).

Do not mix “positive downforce” in a table with “positive C_L up” unless you say so.

This course’s spreadsheets: F_L **and** C_L **up-positive**; report D_w in words or as $|F_L|$ when $F_L < 0$.

From $\Delta P A$ to $F = q C A$

Week 2: if two faces of a part see a pressure difference, the net force has size

$$F \sim \Delta P A$$

ΔP is some fraction of the stream's kinetic-energy density q :

$$\Delta P = C q$$

C is **dimensionless** (Pa/Pa). Substitute:

$$F = C q A$$

Split F into the Week 1 components (drag aft, lift up-positive):

$$F_D = q C_D A, \quad F_L = q C_L A$$

Same chain as a picture

From $\Delta P A$ to the engineering model $F = q C A$

net aero force from a pressure imbalance: $F \sim \Delta P A$

scale the imbalance on q : $\Delta P = C q$ (C dimensionless)

so $F = C q A$

split F into drag and lift: $F_D = q C_D A$, $F_L = q C_L A$

The engineering models

$$F_D = q C_D A, \quad F_L = q C_L A, \quad q = \frac{1}{2} \rho v^2$$

Equivalently:

$$C_D = \frac{F_D}{qA}, \quad C_L = \frac{F_L}{qA}$$

C_D and C_L are dimensionless. They let you compare shapes at different speeds once q and A are known.

Compact form

Coefficients turn a force into a fair number at given q and A

$$F_D = q C_D A$$

$$F_L = q C_L A$$

$$C_D = \frac{F_D}{qA} \quad C_L = \frac{F_L}{qA}$$

C_D , C_L package shape, attitude, and (quietly) Reynolds number.

Anatomy of every symbol

Same pattern for lift: replace D with L .

Anatomy of $F_D = qC_D A$ (same pattern for F_L)

$$F_D = \frac{1}{2} \rho v^2 C_D A$$

the measured (or predicted) drag force

ρ

air density (kg/m^3) — mass of air per volume

v

free-stream speed (m/s)

$$\frac{1}{2} \rho v^2 = q$$

dynamic pressure (Pa) — KE per volume of the stream

A

agreed reference area (m^2) — frontal projection here

C_D

dimensionless leftover: shape, attitude, Re , ...

Units of every symbol

Symbol	Meaning	SI unit
ρ	mass of air per volume	kg/m ³
v	free-stream speed	m/s
$q = \frac{1}{2}\rho v^2$	KE per volume of the stream	Pa
A	agreed reference area (frontal here)	m ²
C_D	leftover shape / attitude / Re factor	dimensionless
F_D	drag force	N

Do not hide a second v^2 inside C_D . Speed already lives in q .

What C_D and C_L "contain"

They are **dimensionless**. They fold in:

- Shape (body, wing, gaps, wheels)
- Attitude (angle of attack, ride height)
- Quietly: Reynolds number and roughness (Week 5)

They do **not** include a hidden extra factor of v^2 — that already sits in q .

If you change only speed and the flow pattern stays similar, C_D is roughly constant and $F_D \propto v^2$.

Stretch: why engineers like C_D

From $F_D = qC_DA$,

$$C_D = \frac{F_D}{qA}$$

Same C_D lets you compare:

- A 1:10 model at 12 m/s to another config at 16 m/s
- (With Re honesty) published car data at different speeds

A raw force in newtons is **not** comparable until you divide by qA .

Reference area A is a convention

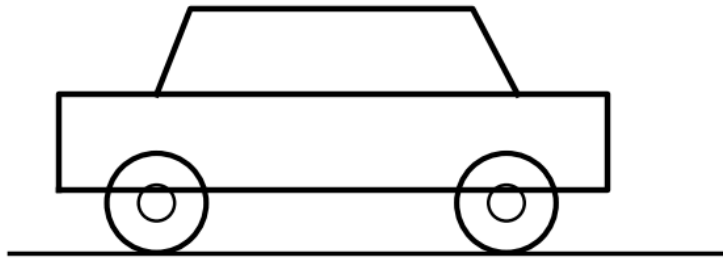
C_D is meaningless without saying which A .

This independent study (default): whole-vehicle frontal (projected) area.

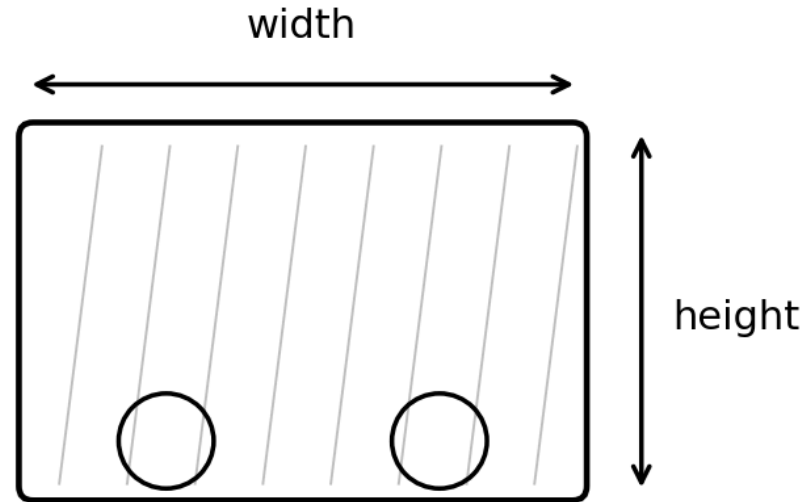
Use it for every full-car config so a wing swap changes C_D, C_L , not A .

Frontal A (the one we freeze)

Course default: one A = vehicle frontal projection



side view



frontal projected area A (hatched)

Do not mix area definitions

Do not mix A definitions in one C_D, C_L table

wing planform A_{wing}



Top view of the wing.
OK for a wing-only study if labeled.

vehicle frontal A



Use this for full-car configs
all term (this course).

If you ever quote a wing-only C_L using planform area, **write** A_{wing} in the table header.

Project rule (write this down)

1. Measure or estimate frontal A of BODY-01 (with wheels, no extra wing if that is your baseline — **say which**).
2. Keep that A for BASE-00, WING-01, SPLIT-02, ...
3. Record A in every CSV (`ref_area_m2` in the lab template).

Changing A mid-term makes Week 9 comparisons garbage.

Worked checkpoint 1 — drag

$$A = 0.015 \text{ m}^2, v = 15 \text{ m/s}, \rho = 1.2 \text{ kg/m}^3, C_D = 0.45.$$

$$q = \frac{1}{2}(1.2)(225) = 135 \text{ Pa}$$

$$F_D = qC_DA = (135)(0.45)(0.015) = 0.911 \text{ N}$$

About 0.91 N aft. That is a small force — exactly the Week 1 sensor-sensitivity warning.

Worked checkpoint 2 — downforce

Same q and A , $C_L = -0.30$.

$$F_L = (135)(-0.30)(0.015) = -0.608 \text{ N}$$

Vertical aero force is 0.61 N **down** ($D_w = 0.61 \text{ N}$).

On the FBD: $N = mg + 0.61 \text{ N}$. For a 12 kg model, that is still a $\sim 0.5\%$ change in N — measure aero force directly if you can.

Worked checkpoint 3 — efficiency

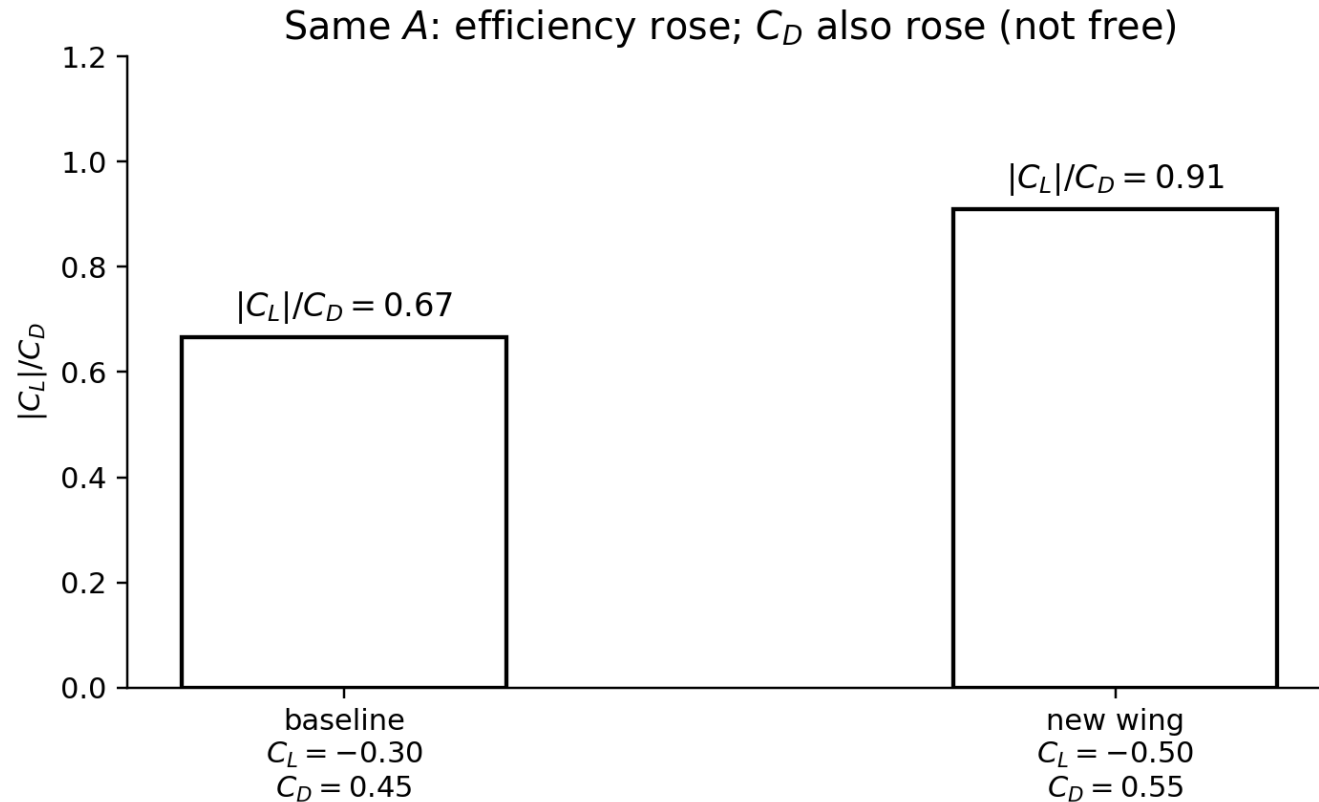
Downforce-to-drag, same A :

$$\mathcal{E} = \frac{|F_L|}{F_D} = \frac{|C_L|}{C_D}$$

Config	C_L	C_D	$\mathcal{E}$
before	−0.30	0.45	$0.30/0.45 = 0.67$
after	−0.50	0.55	$0.50/0.55 = 0.91$

The new wing is **more efficient** by this metric. It is **not** “free”: C_D rose (more drag, more power to hold speed).

Same numbers as a figure



Your Week 1 success metric might be $\mathcal{E}$, or a downforce floor, or a drag cap. **State the rule** (Week 9 will reuse it).

Ballpark C_D (illustrative, not your model)

Object	Typical C_D
Flat plate facing the flow	$\sim 1-2$
Streamlined airfoil (section)	much smaller; Re-dependent
Modern road cars (full vehicle)	often $\sim 0.25-0.35$
Boxy / bluff scale model	often higher than a production car

Relative **changes** when you swap WING-01 still teach design physics even if absolute C_D is "too high."

Estimate *your* model before CAD freeze

Pick A from a sketch or CAD projection.

Assume a trial $C_D \sim 0.4\text{--}0.8$ and C_L from 0 to -0.5 .

Compute F_D and F_L at your Week 2 q values.

If both forces sit below the noise floor of the sensor, **raise v , enlarge A , or change the fixture** (Week 1 / Week 6). Do this on paper now.

Geometric angle of attack

Week 7 is the AoA sweep. This week: **define and record** geometric AoA for WING-01 (chord line vs. oncoming flow or vs. the body reference).

A pretty render with unlabeled incidence is not a physics part.

Design / build tasks this week

- ☐ Written definition of A (how measured, what is included)
- ☐ CAD v1: BODY-01 + at least one interchangeable wing
- ☐ Geometric AoA recorded on the drawing
- ☐ Checkpoint 1–4
- ☐ Optional: predicted C_D , C_L ranges for BASE vs WING-01

Physics checkpoint (full)

1. $q = 135 \text{ Pa}$, $F_D \approx 0.91 \text{ N}$ aft.
2. $F_L \approx -0.61 \text{ N}$ (downforce).
3. $\mathcal{E} : 0.67 \rightarrow 0.91$; after is more efficient, with a drag penalty.
4. Stretch: $C_D = F_D / (qA)$ is dimensionless and comparable across speeds if the flow pattern holds.

Deliverables checklist

Deliverable	Where
Checkpoint solutions	notes or <code>curriculum/week-03-checkpoint.md</code>
Definition of A	project brief or <code>cad/notes/</code>
CAD screenshots + part IDs + AoA	<code>cad/</code>

Due before Week 4. Midterm (Week 5) will ask for predicted C_D , C_L ranges — start the habit now.

How you will be assessed on Week 3

Strong: one frozen A ; correct signs on C_L ; $F = qCA$ arithmetic with units; efficiency discussed as a tradeoff.

Weak: C_D quoted with no A ; "the wing adds 30% downforce" with no q ; mixing planform and frontal area.

Looking ahead — Week 4

Bernoulli and continuity — **and their limits.**

You will write honest captions for the diffuser and the wing: marketing claim vs. physics claim. Separation and wakes enter the vocabulary. Do not wait until then to keep ΔP and q honest.

Instructor talking points

- Freeze A in writing this week; do not renegotiate every CAD revision.
- If the student reports only newtons, ask for $C_D = F_D / (qA)$.
- Efficiency is not the only goal — a design brief might demand a D_w floor.
- Preview Re (Week 5) if they treat a model C_D as a street-car spec.

References for this lecture

- Course note: [physics/03-drag-and-lift.md](#)
- Week plan: [curriculum/week-03-drag-lift-coefficients.md](#)
- q recap: [physics/02-pressure-and-dynamic-pressure.md](#)
- Lab CSV columns: [labs/data-template.csv](#)
- Figures: [curriculum/slides/figures/](#)

End of Week 3 lecture

Next actions

1. Finish checkpoint
2. Write the A definition
3. CAD v1 + AoA
4. Push to the course GitHub repo

Questions?